

numerical representations back into words and ensuring that the response is grammatically correct.

“These steps necessitate a significant amount of computing power, and ChatGPT relies on high-performance hardware to ensure that it can generate responses quickly and accurately. It generally requires powerful graphics processing units (GPUs) or tensor processing units (TPUs) to handle the massive amounts of data involved in natural language processing,” he explains.

Platforms like Microsoft Azure and IBM Cloud come to the rescue, providing a seamless fabric of compute power required to train AI models. According to reports, ChatGPT initially used a Microsoft Azure supercomputing infrastructure, powered by Nvidia GPUs, which Microsoft built specifically for OpenAI. This year, after Microsoft announced its multi-year investment in OpenAI, it also phenomenally upgraded OpenAI’s infrastructure.

Sachindra Joshi, IBM distinguished engineer, Conversational Platforms, IBM Research India, remarks that, “Building and deploying foundation models requires proper infrastructure. Training models like ChatGPT requires thousands of GPUs, making it extremely difficult for many entities to train these kinds of models from scratch. Deploying these models for inference is also very resource consuming and a lot of research is happening around better learning models with fewer parameters, and making inference costs cheaper.”

With great power comes great responsibility

All that impressive technology working in the background gives you not just a lovable chatbot that you can chat and while away your time with, but also a very capable API that can be used to enhance applications and business platforms.

Is ChatGPT open source

Despite the company being called OpenAI, the code of ChatGPT is not open sourced. Since the company was not available for comment, we asked ChatGPT itself about it, and here is what it had to say:

“As an AI language model, my code is not open sourced. However, the underlying technology that powers me is based on research and development by OpenAI, and some of the tools and libraries used to create me are open source. OpenAI has also released several pre-trained models for use in natural language processing tasks, which are available for free download and use. If you’re interested in exploring these resources, you can find more information on the OpenAI website.”

This is something that has miffed many in the open source world, because ChatGPT uses quite a lot of open source components and open knowledge as well. (Practically most of its training has happened on information garnered from the Web, contributed by people from around the world.) Perhaps it is not giving back as much as it takes?

ChatGPT can converse on varied subjects, draft emails, make lists, translate text into different languages, summarise long texts, assist in various domains like education, healthcare, finance, and customer service, and do much, much more. It can adapt to different communication channels and be integrated with external data sources and application programming interfaces (APIs) to provide more customised and comprehensive responses.

Hand-in-hand with all this power comes a lot of risk as well. From privacy and plagiarism to cybersecurity issues, and the very serious possibility of it dumbing down the human brain, there are several issues that we must be aware of, so we can make responsible use of generative AI.

Indeed, ChatGPT has taken trendwatchers by surprise, by racing to the peak of the hype cycle within a phenomenally short span of time. Unlike the metaverse, which people are still struggling to understand, let alone adopt, generative AI (especially ChatGPT) has already proved its worth to people and seems to be here to stay.

“ChatGPT has revolutionised the accessibility of AI to the masses, providing people with the opportunity to interact with it in ways that were once exclusive. The pace at which enterprises are investing in AI is remarkable, having surged by 2.5 times over the last five years. Yet, the number of AI products available today is several multiples higher than it was in 2017. The impact of this disruption is truly remarkable,” says Sahay.

He explains that, in the near future, these generative AI models will become better, less expensive, and more accurate. Because these models are becoming multimodal, a single model will soon be able to synthesise information from all four modalities—text, image, video, as well as speech.

Sahay signs off saying, “This is a critical future element that everyone is anticipating, and will be more closely monitored and moderated for ethical reasons. There is no doubt that everyone is on board to ensure that both privacy and accountability are addressed, and that more regulation and governance in this area are clearly needed.” **END** 

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